

Exercise 1: First Normal Form (1NF) - Answers

- The `Course`, `Course_Grade` and `Instructor` columns violate **1NF**. This is because they contain multiple, non-atomic values (e.g., "Math, Physics" and "A, B") in a single cell, which is a violation of the first rule of 1NF. Similarly, the `Instructor` column also violates 1NF as it lists multiple instructors.
- All students should have unique student IDs. In this case, "Bob" is associated with 2 students IDs – 102 and 108. We can assume that because we only have the first name, the surname of the students 102 and 108 will be different thus they are two different students. Or this could also be a mistake and there is only 1 Bob. In this case, we again need the full name and to check then the students IDs. If the names are duplicates, we can assume in this simple case there is a mistake with the IDs. The solution will be to edit the student ID.
- To normalize this table to 1NF, we need to ensure each cell contains only one value. We can achieve this by creating a new row for each course a student is enrolled in.

Student ID	Name	Course	Instructor	Course Grade
100	Alice	Maths	Smith, Jones	A
100	Alice	Physics	Doe, John	B
102	Bob	Physics	Doe, John	A
104	Alice	English	Doe Jane	A
104	Alice	Art	Doe Jane	A
108	Bob	Art	James, Joe	B
108	Bob	Maths	Doe, Jay	B

Or

Student ID	Full Name	Course	Instructor	Course Grade
100	Alice	Maths	Smith, Jones	A
100	Alice	Physics	Doe, John	B
102	Bob Martin	Physics	Doe, John	A
104	Alice	English	Doe Jane	A
104	Alice	Art	Doe Jane	A
102	Bob Martin	Art	James, Joe	B
102	Bob Martin	Maths	Doe, Jay	B

Exercise 2: Second Normal Form (2NF) – Answers

1. The **primary keys** are (OrderID, ProductID).
2. The columns OrderDate and CustomerName are **functionally dependent** on only a part of the primary key, specifically OrderID. The ProductName and Price columns are functionally dependent on only ProductID.
3. The data redundancy and update anomalies are:
 - 3.1. **Data Redundancy:** The OrderDate and CustomerName are repeated for every product in a single order (e.g., Order 1001's details are repeated for both the Laptop and the Mouse). Similarly, the ProductName and Price are repeated for every order that contains that product (e.g., the Laptop's details are listed for both orders 1001 and 1002).
 - 3.2. **Update Anomalies:** If the CustomerName for Order 1001 needed to be changed, we would have to update multiple rows. If we miss one, the data becomes inconsistent.
4. To achieve 2NF, we decompose the table by separating the attributes that have partial dependencies into new tables.

Orders Table

OrderID	OrderDate	CustomerName

OrderItems Table

OrderID	ProductID	Quantity

Products Table

ProductID	ProductName	Price

Exercise 3: Third Normal Form (3NF) - Answers

1. The primary key is **EmployeeID**.
2. The transitive dependency is: **EmployeeID -> DepartmentID -> (DepartmentName, DepartmentLocation)**. The non-key attributes DepartmentName and DepartmentLocation are dependent on the non-key attribute DepartmentID, rather than directly on the primary key EmployeeID.
3. The potential issues are:
 - **Insertion Anomaly:** We cannot add a new department to the table until an employee is assigned to it.
 - **Update Anomaly:** If the DepartmentLocation for "Sales" changes from "New York" to "Boston", we would have to update multiple rows. If we miss one, the data becomes inconsistent.
 - **Deletion Anomaly:** If employee E001 (Alice) is deleted, we lose all information about the "Sales" department (unless another employee is in that department).
4. To achieve 3NF, we remove the transitive dependency by creating a new table for the Department information.

Employees Table

EmployeeID	DepartmentID	EmployeeName

Departments Table

DepartmentID	DepartmentName	DepartmentLocation